

Assignment

Digital Electronics

Q1. convert:

(i)

$$(a) (85.63)_{10} \rightarrow (X)_2$$

$$\begin{array}{r|l} 2 & 85 \end{array}$$

$$\begin{array}{r|ll} 2 & 42 & 1 \end{array}$$

$$\begin{array}{r|ll} 2 & 21 & 0 \end{array}$$

$$\begin{array}{r|ll} 2 & 10 & 1 \end{array}$$

$$\begin{array}{r|ll} 2 & 5 & 0 \end{array}$$

$$\begin{array}{r|ll} 2 & 2 & 1 \end{array}$$

$$\begin{array}{r|ll} 2 & 1 & 0 \end{array}$$

$$\begin{array}{r|ll} & 0 & 1 \end{array}$$

$$(85)_{10} = (1010101)_2$$

now, 0.63

denimal

$$0.63 \times 2 = 1.26$$

1

$$0.26 \times 2 = 0.52$$

0

$$0.52 \times 2 = 1.04$$

1

$$0.04 \times 2 = 0.08$$

0

$$0.08 \times 2 = 0.16$$

0

$$0.16 \times 2 = 0.32$$

0

$$0.32 \times 2 = 0.64$$

0

$$0.64 \times 2 = 1.28$$

1

$$(0.63)_{10} = (0.10100001)_2$$

(approx)

$$\text{hence } (85.63)_{10} = (1010101.10100001)_2$$

$$X = 1010101.10100001$$

$$(b) (300.45)_{10} = (X)_8$$

$$\begin{array}{r|l} 8 & 300 \end{array}$$

$$\begin{array}{r|ll} 8 & 37 & 4 \end{array}$$

$$\begin{array}{r|ll} 8 & 4 & 5 \end{array}$$

$$\begin{array}{r|ll} & 0 & 4 \end{array}$$

$$(300)_{10} = (454)_8$$

now. 45

decimal

$$\cdot 45 \times 8 = 3.60$$

3

$$\cdot 60 \times 8 = 4.80$$

4

$$\cdot 80 \times 8 = 6.40$$

6

$$\cdot 40 \times 8 = 3.20$$

3

$$\cdot 20 \times 8 = 1.60$$

1

$$\Rightarrow (.45)_{10} = (.34631)_8$$

$$\text{Hence } (300.45)_{10} = (454.34631)_8$$

$$\Rightarrow X = 454.34631_{-}$$

$$(c) (F3A7C2)_{16} = (X)_2$$

Hexadecimal :-

In hexadecimal

$$F = 1111$$

$$3 = 0011$$

$$A = 1010$$

$$7 = 0111$$

$$C = 1100$$

$$2 = 0010$$

$$0 \ 0000 \quad A \ 1010$$

$$1 \ 0001 \quad B \ 1011$$

$$2 \ 0010 \quad C \ 1100$$

$$3 \ 0011 \quad D \ 1101$$

$$4 \ 0100 \quad E \ 1110$$

$$5 \ 0101 \quad F \ 1111$$

$$6 \ 0110$$

$$7 \ 0111$$

$$8 \ 1000$$

$$9 \ 1001$$

$$\Rightarrow (F3A7C2)_{16} = (1111 \ 0011 \ 1010 \ 0111 \ 1100 \ 0010)_2$$

$$(d) (1101.101)_2 = (X)_{10}$$

$$\begin{aligned} \text{First } (1101)_2 &= (1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0)_{10} \\ &= (8 + 4 + 1)_{10} = (13)_{10} \end{aligned}$$

$$\begin{aligned} \text{Then } (.101) &= (1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3})_{10} \\ &= (0.5 + 0 + 0.125) = (0.625)_{10} \end{aligned}$$

$$\Rightarrow (1101.101) = (13.625)_{10}$$

$$\Rightarrow X = 13.625$$

$$(e) (1101010)_2 = (X)_{\text{gray}}$$

gray code: First bit same, then each bit = XOR of previous and current bit

$$\text{Binary} = \underline{1} \ \underline{1} \ \underline{0} \ \underline{1} \ \underline{0} \ \underline{1} \ \underline{0}$$

$$\text{First bit} = \underline{1}$$

$$\text{2nd} \quad 1 \oplus 1 = 0$$

$$\text{3rd} \quad 0 \oplus 1 = 1$$

$$1 \oplus 0 = 1$$

$$0 \oplus 1 = 1$$

$$1 \oplus 0 = 1$$

$$0 \oplus 1 = 1$$

$$\Rightarrow \text{gray code of } (1101010)_2 = (1011111)_{\text{gray}}$$

$$\Rightarrow X = 1011111$$

(ii)(a) 27

$$2 \mid 27$$

$$2 \mid 13 \ 1$$

$$2 \mid 6 \ 1$$

$$2 \mid 3 \ 0$$

$$2 \mid 1 \ \emptyset$$

$$0 \ 1$$

$$\rightarrow \text{Binary} = (1\emptyset011)_2$$

$$8 \text{ bit form} = (00010011)_2$$

$$\rightarrow \text{ASCII}$$

$$2 = 50 \text{ in ASCII}$$

$$7 = 55 \text{ in ASCII}$$

$$\text{Binary: } 1\emptyset011$$

$$\text{First bit} = 1$$

$$\text{2nd} = 1 \oplus \emptyset = 0$$

$$\text{3rd} = 0 \oplus 1 = 1$$

$$4 = 1 \oplus 0 = 1$$

$$5 = 1 \oplus 1 = 0$$

$$\text{gray} = 10010$$

Excess 3

$$2 + 3 = 5 = (0101)_2$$

$$7 + 3 = 10 = (1010)_2$$

$$\text{Excess-3 code: } (0101 \ 1010)$$

b) 396

→ Binary

2	396	
2	198	0
2	99	0
2	49	1
2	24	1
2	12	0
2	6	0
2	3	0
2	1	1
	0	1

Binary = $(110001100)_2$

→ ASCII

3 = 51 in ASCII

9 = 57 in ASCII

6 = 54 in ASCII

→ Gray code

Binary:- 110001100

1st bit = 1

2 = $1 \oplus 1 = 0$

3 = $0 \oplus 1 = 1$

4 = $0 \oplus 0 = 0$

5 = $0 \oplus 0 = 0$

6 = $1 \oplus 0 = 1$

7 = $1 \oplus 1 = 0$

8 = $0 \oplus 1 = 1$

9 = $0 \oplus 0 = 0$

Gray code: 101001010

1-2-3-4-5-6-7-8-9-10-11-12-13-14-15-16-17-18-19-20-21-22-23-24-25-26-27-28-29-30-31-32-33-34-35-36-37-38-39-40-41-42-43-44-45-46-47-48-49-50-51-52-53-54-55-56-57-58-59-60-61-62-63-64-65-66-67-68-69-70-71-72-73-74-75-76-77-78-79-80-81-82-83-84-85-86-87-88-89-90-91-92-93-94-95-96-97-98-99-100-101-102-103-104-105-106-107-108-109-110-111-112-113-114-115-116-117-118-119-120-121-122-123-124-125-126-127-128-129-130-131-132-133-134-135-136-137-138-139-140-141-142-143-144-145-146-147-148-149-150-151-152-153-154-155-156-157-158-159-160-161-162-163-164-165-166-167-168-169-170-171-172-173-174-175-176-177-178-179-180-181-182-183-184-185-186-187-188-189-190-191-192-193-194-195-196-197-198-199-200-201-202-203-204-205-206-207-208-209-210-211-212-213-214-215-216-217-218-219-220-221-222-223-224-225-226-227-228-229-230-231-232-233-234-235-236-237-238-239-240-241-242-243-244-245-246-247-248-249-250-251-252-253-254-255-256-257-258-259-260-261-262-263-264-265-266-267-268-269-270-271-272-273-274-275-276-277-278-279-280-281-282-283-284-285-286-287-288-289-290-291-292-293-294-295-296-297-298-299-300-301-302-303-304-305-306-307-308-309-310-311-312-313-314-315-316-317-318-319-320-321-322-323-324-325-326-327-328-329-330-331-332-333-334-335-336-337-338-339-340-341-342-343-344-345-346-347-348-349-350-351-352-353-354-355-356-357-358-359-360-361-362-363-364-365-366-367-368-369-370-371-372-373-374-375-376-377-378-379-380-381-382-383-384-385-386-387-388-389-390-391-392-393-394-395-396-397-398-399-400-401-402-403-404-405-406-407-408-409-410-411-412-413-414-415-416-417-418-419-420-421-422-423-424-425-426-427-428-429-430-431-432-433-434-435-436-437-438-439-440-441-442-443-444-445-446-447-448-449-450-451-452-453-454-455-456-457-458-459-460-461-462-463-464-465-466-467-468-469-470-471-472-473-474-475-476-477-478-479-480-481-482-483-484-485-486-487-488-489-490-491-492-493-494-495-496-497-498-499-500-501-502-503-504-505-506-507-508-509-510-511-512-513-514-515-516-517-518-519-520-521-522-523-524-525-526-527-528-529-530-531-532-533-534-535-536-537-538-539-540-541-542-543-544-545-546-547-548-549-550-551-552-553-554-555-556-557-558-559-560-561-562-563-564-565-566-567-568-569-570-571-572-573-574-575-576-577-578-579-580-581-582-583-584-585-586-587-588-589-590-591-592-593-594-595-596-597-598-599-600-601-602-603-604-605-606-607-608-609-610-611-612-613-614-615-616-617-618-619-620-621-622-623-624-625-626-627-628-629-630-631-632-633-634-635-636-637-638-639-640-641-642-643-644-645-646-647-648-649-650-651-652-653-654-655-656-657-658-659-660-661-662-663-664-665-666-667-668-669-670-671-672-673-674-675-676-677-678-679-680-681-682-683-684-685-686-687-688-689-690-691-692-693-694-695-696-697-698-699-700-701-702-703-704-705-706-707-708-709-710-711-712-713-714-715-716-717-718-719-720-721-722-723-724-725-726-727-728-729-730-731-732-733-734-735-736-737-738-739-740-741-742-743-744-745-746-747-748-749-750-751-752-753-754-755-756-757-758-759-760-761-762-763-764-765-766-767-768-769-770-771-772-773-774-775-776-777-778-779-780-781-782-783-784-785-786-787-788-789-790-791-792-793-794-795-796-797-798-799-800-801-802-803-804-805-806-807-808-809-810-811-812-813-814-815-816-817-818-819-820-821-822-823-824-825-826-827-828-829-830-831-832-833-834-835-836-837-838-839-840-841-842-843-844-845-846-847-848-849-850-851-852-853-854-855-856-857-858-859-860-861-862-863-864-865-866-867-868-869-870-871-872-873-874-875-876-877-878-879-880-881-882-883-884-885-886-887-888-889-890-891-892-893-894-895-896-897-898-899-900-901-902-903-904-905-906-907-908-909-910-911-912-913-914-915-916-917-918-919-920-921-922-923-924-925-926-927-928-929-930-931-932-933-934-935-936-937-938-939-940-941-942-943-944-945-946-947-948-949-950-951-952-953-954-955-956-957-958-959-960-961-962-963-964-965-966-967-968-969-970-971-972-973-974-975-976-977-978-979-980-981-982-983-984-985-986-987-988-989-990-991-992-993-994-995-996-997-998-999-1000-1001-1002-1003-1004-1005-1006-1007-1008-1009-1010-1011-1012-1013-1014-1015-1016-1017-1018-1019-1020-1021-1022-1023-1024-1025-1026-1027-1028-1029-1030-1031-1032-1033-1034-1035-1036-1037-1038-1039-1040-1

Leaves of Verbena stricta

DATE: 11/11/2019

Feb 17 1909

→ Exien 3 = (0110 1100 1001)

transm. brdf. + (transm. loss) %

• 2005 was an eventful year for the firm.

[illegible]

2

$$(21.8 \pm 0.2 \text{ CF}) = \text{salmon p. mad}$$

$$4+3 = 7 = (0111)$$

→ 3. wrotanego bitowania $0+3=3 \Rightarrow (0011)_{(4)}$

$$9+3 = 12 = (1100)$$

$$6+3 = 9 = (1001)$$

$$\Rightarrow (0111 \ 0011 \ 1100 \ 1001)$$

[illegible]

1000

$$(1e11) = a_1(2b)$$

$$(0|0|0|) = (0^4)$$

01-0000-0000

→ gray 1 0000 0000 0000

→ 1100 0000 0000

(iii) $3^2 = 9$

simple procedure to convert base 3 to base 9.

Procedure:

1. Pair the number in groups of 2 since $3^2 = 9$ from right side
2. For each pair calculate 9 representation as $3 \times (\text{1st element}) + \text{2nd element}$
3. Write the 9 representations in same order.

Applying to $(2110201102220112)_3$

1. grouping: 21 10 20 11 02 22 01 12

$$\text{Value} = \begin{array}{cccccccc} 2(2)+1 & 3(1)+0 & 3(2) & 3(1)+1 & 3(0)+2 & 3(2)+2 & +1 & 3(1)+2 \\ 7 & 3 & 6 & 4 & 2 & 8 & 1 & 5 \end{array}$$

Base 9 number = $(73642815)_9$

(iv) Binary arithmetic operations :-

a) $+42 + (-13)$

$$\begin{array}{r|l} 2 & 42 \\ \hline 2 & 21 \quad 0 \\ 2 & 10 \quad 1 \\ 2 & 5 \quad 0 \\ 2 & 2 \quad 1 \\ 2 & 1 \quad 0 \\ \hline & 0 \quad 1 \end{array}$$

$$\begin{array}{r|ll} 2 & 13 & \\ \hline 2 & 6 & 1 \\ 2 & 3 & 0 \\ 2 & 1 & 1 \\ & 0 & 1 \end{array}$$

$$\begin{aligned} (42)_{10} &= (101010)_2 \\ &= (00101010)_2 \end{aligned}$$

$$\begin{aligned} (13)_{10} &= (1101)_2 \\ &= (00001101)_2 \end{aligned}$$

since -13 is -ve, 1's complement: 11110010

2's complement: $\begin{array}{r} 11110010 \\ +1 \\ \hline 11110011 \end{array}$

addition: 00101010

$\begin{array}{r} 11110011 \\ +00101010 \\ \hline 10001101 \end{array}$

discarded

Result: $(0001101)_2$

b) $(-42) - (-13)$
 $(-42) + (+13)$

$(13)_{10} = (0001101)_2$

$(42)_{10} = (00101010)_2$

since -42 is -ve: 1's complement: 11010101

2's complement: $\begin{array}{r} 11010101 \\ +1 \\ \hline 11010110 \end{array}$

addition: 11010110

$\begin{array}{r} 11010110 \\ +00001101 \\ \hline 11100011 \end{array}$

Result: $(11100011)_2$

Q2. (a) system has two identical circuits in parallel
say we have X and Y outputs
when circuits are good $X=Y$
when error, $X \neq Y$ (sometimes)

failure detection method:

Feed the outputs to an X-OR gate

say $F = X \oplus Y$

when $X=Y \Rightarrow F=0$ (normal)

but when $X \neq Y \Rightarrow F=1$ (error detected).

b) Boolean reductions

$$(i) A'C' + A*BC + AC$$

$$= C'(A'+A) + A*BC$$

$$= C' + ABC$$

Identity $X + X'Y = X + Y$

$$\Rightarrow C' + A*BC = C' + A(B)(C) = C' + AB$$

$$= \underline{C' + AB} \quad \text{Three literals.}$$

$$(ii) A'B(D'+C'D) + B(A+A'CD)$$

here $D'+C'D = D'+C'$ using Identity $X+X'Y=X+Y$

also $A+A'CD = A+CD$

$$\Rightarrow \text{exp} = A'B(D'+C') + B(A+CD)$$

$$= B(A'D' + A'C' + A + CD)$$

$$= B(A + A'C' + A'D' + CD)$$

$$= B(A + C' + D' + CD) \quad \text{using same Identity}$$

$$= B(A + \underline{C' + CD} + D')$$

$$= B(A + C' + D + D')$$

$$= B((A + D') + (C' + D))$$

$$= B(A + C' + 1)$$

$$= B(1)$$

$$= \underline{B} \quad \text{one Literal}$$

$$(iii) (A'+C)(A'+C') (A+B+C'D)$$

$$(\cancel{A'A} + \cancel{A'C'} + \cancel{CA'} + \cancel{CC'}) (\cancel{A+B+C'D})$$

~~Q3~~ A

$$= A'(\cancel{A'C} + \cancel{CC'}) (\cancel{A+B+C'D})$$

$$= \cancel{A'(A'C)} (\cancel{A+B+C'D})$$

using law $(X+Y)(X+Z) = X+YZ$

$$= (A' + CC') (A+B+C'D)$$

$$= A' (A+B+C'D)$$

$$= A'A + A'B + AC'D$$

$$= \underline{A'B + AC'D}$$

4 literals

Q3. Convert of standard SOP and POS forms

$$(i) (AB+C)(B+C'D)$$

$$F = \cancel{A}BB(B+C'D) + C(B+C'D)$$

$$= \cancel{A}BB + \cancel{A}BC'D + CB + CC'D$$

$$\left. \begin{array}{l} \text{here } \cancel{A}BB = AB \\ CC'D = 0 \end{array} \right\}$$

$$\Rightarrow F = AB + ABC'D + CB$$

$$= AB + BC + ABC'D$$

$$\text{since } X + XY = X$$

$$\Rightarrow AB + ABC'D = AB$$

$$\Rightarrow \underline{F = AB + BC}$$

Canonical SOP (minterms)

and POS (maxterms)

A	B	C	D	AB	BC	F	minterm / maxterm
0	0	0	0	0	0	0	M ₀
0	0	0	1	0	0	0	M ₁
0	0	1	0	0	0	0	M ₂
0	0	1	1	0	0	0	M ₃
0	1	0	0	0	0	0	M ₄
0	1	0	1	0	0	0	M ₅
0	1	1	0	0	1	1	M ₆
0	1	1	1	0	1	1	M ₇
1	0	0	0	0	0	0	M ₈
1	0	0	1	0	0	0	M ₉
1	0	1	0	0	0	0	M ₁₀
1	0	1	1	0	0	0	M ₁₁
1	1	0	0	1	0	0	M ₁₂
1	1	0	1	1	0	1	M ₁₃
1	1	1	0	1	1	1	M ₁₄
1	1	1	1	1	1	1	M ₁₅

from $F=1$ SOP = $\sum m(6,7,12,13,14,15)$

$F=0$ POS = $\prod M(0,1,2,3,4,5,8,9,10,11)$

$$(ii) = X' + X(X + Y')(Y + Z')$$

$$F = X' + X(1)(Y + Z')$$

$$= X' + X(Y + Z')$$

$$= X' + XY + XZ'$$

$$= \underline{X' + XZ'} + XY$$

$$= X' + Z' + XY \quad (\text{as } X + X'Y = X + Y)$$

$$= X' + XY + Z'$$

$$= X' + Y + Z' \quad (\text{as } X + X'Y = X + Y)$$

X	Y	Z	X'	Y'	Z'	$F = X' + Y + Z'$	Minterm	Maxterm
0	0	0	1	0	1	1	m_0	-
0	0	1	1	0	0	1	m_1	-
0	1	0	1	1	1	1	m_2	-
0	1	1	1	1	0	1	m_3	-
1	0	0	0	0	1	1	m_4	-
1	0	1	0	0	0	0	-	M_5
1	1	0	0	1	1	1	m_6	-
1	1	1	0	1	0	1	m_7	-

$\therefore$ Canonical SOP = $\sum m(0, 1, 2, 3, 4, 6, 7)$
 POS = $\prod M(5)$

(ii) $B'A + A'B + B'A$

$= (B' + A' + B)$

$= (A' + (B + B'))$

$= (A' + 1)$ (as $B + B' = 1$)

$= (1) = 1$

A	B	A	B'	A'	B'A	A'B	BA	F	Min	Max
0	0	0	1	1	0	0	0	0	m_0	M_0
0	0	1	1	1	1	1	0	1	m_1	
0	1	0	0	1	0	0	0	0		M_2
0	1	1	0	1	0	1	1	1	m_3	
1	0	0	1	0	0	0	0	0		M_4
1	0	1	1	0	1	0	0	1	m_5	
1	1	0	0	0	0	0	0	0		M_6
1	1	1	0	0	0	0	1	1	m_7	

$\therefore$ Canonical SOP = $\sum m(1, 3, 5, 7)$
 POS = $\prod M(0, 2, 4, 6)$

$$\begin{aligned}
 \text{(iv)} \quad & (Xy+z)(xz+y) \\
 & xy(xz+y) + z(xz+y) \\
 & xyxz + xy + zxz + zy \\
 & xyz + xy + xz + yz \\
 & (xy + xyz) + xz + yz \\
 & = xy + xz + yz \quad [\text{as } xy + xyz = xy]
 \end{aligned}$$

X	y	z	xy	xz	yz	F	Minterm	Maxterm
0	0	0	0	0	0	0	-	M ₀
0	0	1	0	0	0	0	-	M ₁
0	1	0	0	0	0	0	-	M ₂
0	1	1	0	0	1	1	m ₃	
1	0	0	0	0	0	0	-	M ₄
1	0	1	0	1	1	1	m ₅	
1	1	0	1	0	1	1	m ₆	
1	1	1	1	1	1	1	m ₇	

$$\text{Canonical SOP} = \sum m(3, 5, 6, 7)$$

$$\text{POS} = \prod M(0, 1, 2, 4)$$

Simplify using k-maps:-

$$(i) \quad F(x, y, z) = \sum m(1, 4, 5, 6)$$

minterms in binary:

$$1 = 001$$

$$4 = 100$$

$$5 = 101$$

$$6 = 110$$

k-map:

W \ XY	$\bar{X}Y$	$\bar{X}\bar{Y}$	XY	$X\bar{Y}$
$\bar{W}$	0	1	3	2
W	4	5	7	6

from the kmap $\Rightarrow F = \bar{X}W + \bar{X}Y + W\bar{Y}$

(iii) $F(A, B, C, D, E) = \sum_m (0, 2, 3, 4, 5, 6, 7, 11, 15, 16, 18, 19, 23, 27, 31)$

kmap ($E=0$)

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	2	3
$\bar{A}B$	4	5	6	7
$A\bar{B}$	8	9	10	11
AB	12	13	14	15

$E=1$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	2	3	4
$\bar{A}B$	5	6	7	8
$A\bar{B}$	9	10	11	12
AB	13	14	15	16

$$F = DE + \bar{A}\bar{B}C + \bar{B}\bar{C}E$$

(iv) $F(A, B, C, D) = \Pi_M(0, 1, 2, 3, 4, 10, 11) \Rightarrow$ minterms (-
 $2, 5, 7, 6, 12, 13, 15,$
 $14, 8, 9$

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$\overline{AB} = 11$

$$F = AB + AC' + A'BC + A'BD$$

$$= AB + AC' + A'B(C+D)$$

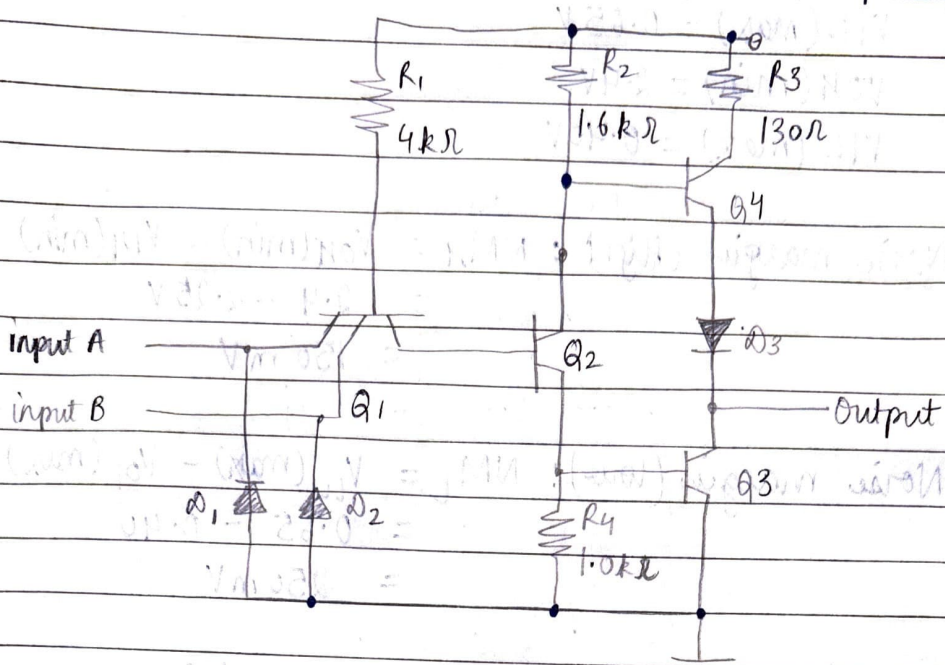
(ii) $A^2B'C + B'C'D + BCD + AC'D' + A'B'C + A'BC'D$

This expression yields minterms = 1, 2, 3, 5, 7, 9, 10, 14, 15

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$$F = A'D + AC + B'C + B'D$$

Q4. (a) Draw a TTL circuit with totem pole output



TTL NAND gate circuit

verification:-

Case I: $A=0$ (low), $B=0$ (low)
 $\Rightarrow$ Q1 (forward bias) $\Rightarrow$ Q2 base LOW $\Rightarrow$ Q2 off $\Rightarrow$ Q4 off
 pull out high through Q3
 output = 1 (matches NAND)

Case II: $A=0$, $B=1$ [and case III: $A=1$, $B=0$]
 Q1 conducts emitter tied to $A=0$, $B \rightarrow$ pulled low $\Rightarrow$
 Q2 off $\Rightarrow$ Q4 off $\Rightarrow$ Q3 ON
 output = 1 (matches NAND)

Case IV: $A=1$, $B=1$
 Q1 = high $\Rightarrow$ Q1 off $\Rightarrow$ B = high $\Rightarrow$ Q2 = on $\Rightarrow$ Q4 on
 $\Rightarrow$ Q3 driven off
 output = 0 (matches NAND)

(b)

$$V_{IH}(\min) = 2.25 \text{ V}$$

$$V_{IL}(\max) = 0.65 \text{ V}$$

$$V_{OH}(\min) = 2.4 \text{ V}$$

$$V_{OL}(\max) = 0.40 \text{ V}$$

Noise margin (High): $NM_H = V_{OH}(\min) - V_{IH}(\min)$

$$= 2.4 - 2.25 \text{ V}$$

$$= 150 \text{ mV}$$

Noise margin (low): $NM_L = V_{IL}(\max) - V_{OL}(\max)$

$$= 0.65 - 0.40$$

$$= 250 \text{ mV}$$

NM_H is 0.15 V and NM_L is 0.25 V which are very small value

$\therefore$ Hence, the high and low noise margins are not suitable for a circuit design. ~~as not~~